Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the validation accuracy during training.

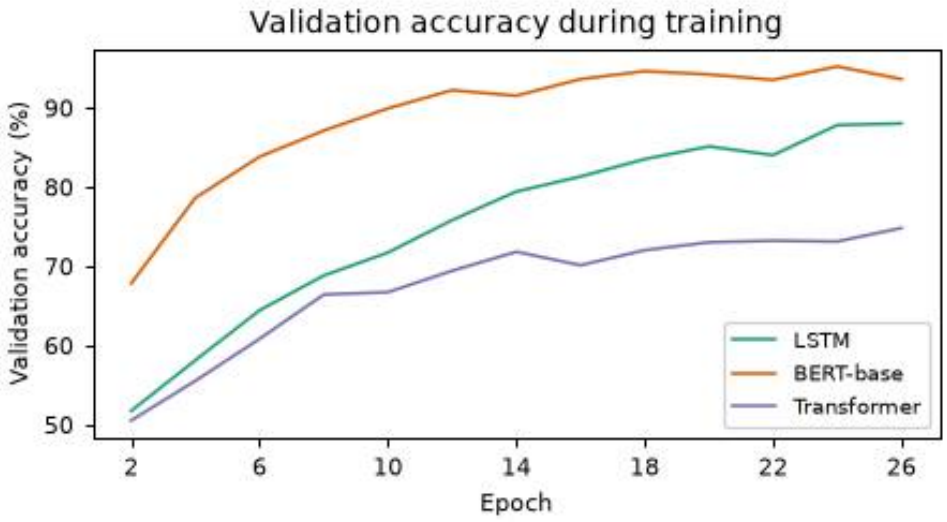


Figure 1: Validation accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 26, LSTM achieves over 85% validation accuracy. Transformer attains a validation accuracy of 87.8% at epoch 24 (Figure 1). Transformer attains a validation accuracy of 60.3% at epoch 16 (Figure 1). LSTM attains a validation accuracy of 64% at epoch 6.

Figure 2 shows the test accuracy during training.

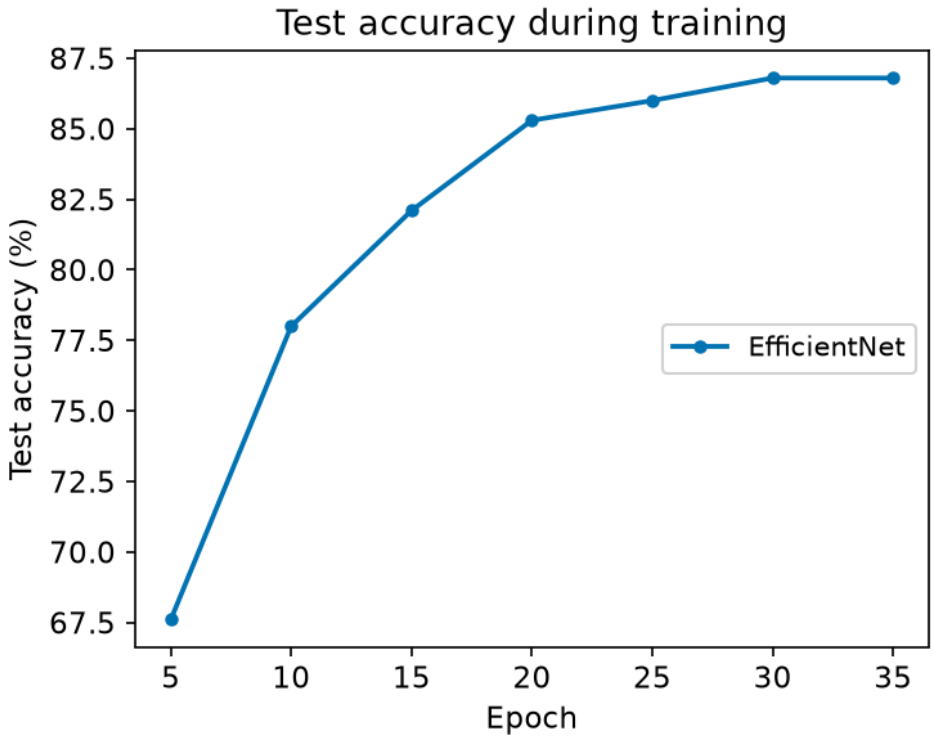


Figure 2: Test accuracy during training for the models in this experiment.

EfficientNet scores 9.9 points higher on 30 than on 10.

Figure 3 shows the test accuracy during training.

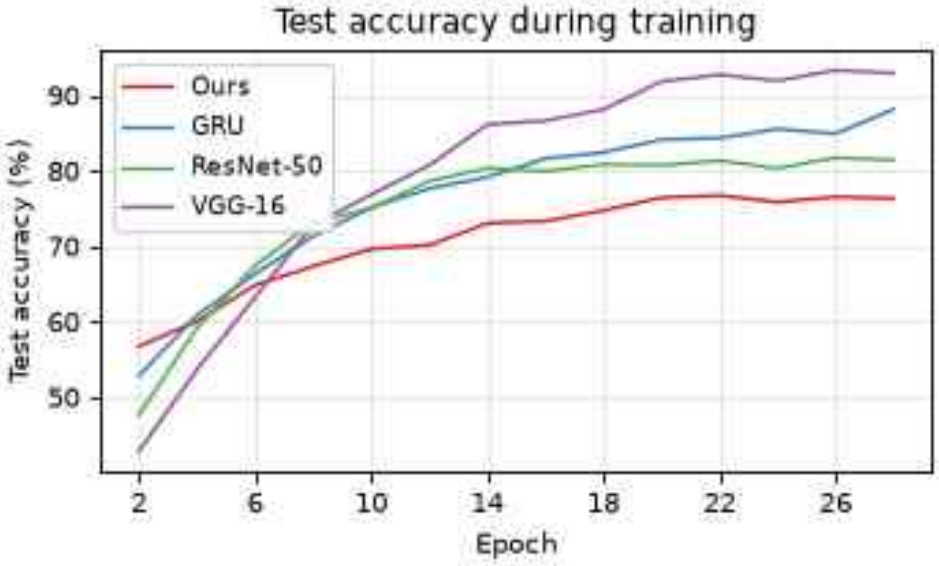


Figure 3: Test accuracy during training for the models in this experiment.

ResNet-50 reaches 74.3% test accuracy at epoch 16. ResNet-50 reaches 81.4% test accuracy at epoch 24. At epoch 18, VGG-16 is about 7 points better than ResNet-50.

Figure 4 shows the training loss.

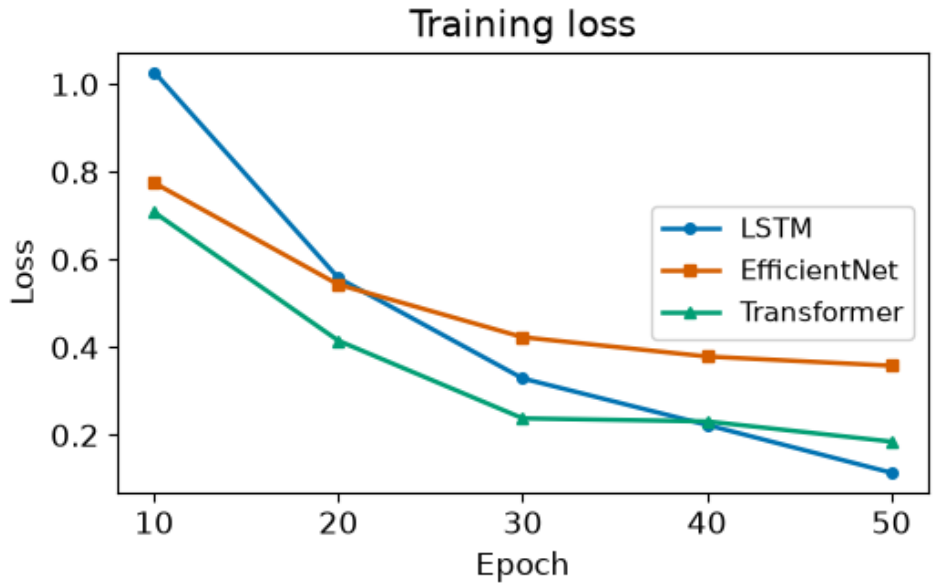


Figure 4: Training loss for the models in this experiment.

EfficientNet has a loss of 0.278 at epoch 40. LSTM has a loss of over 0.3 at epoch 30.

3 Discussion

XGBoost reaches 93.1% accuracy on Tiny-ImageNet. Overall, the results are in line with what we expected from the literature.